

DRIFT-SENSE BENCHMARK REPORT

DRAM Wafer Alignment Evaluation: Multi-Threshold Metrics & Failure Analysis

Executive Overview: Official live evaluation of the Drift-Sense SEM navigation engine on **DRAM High-Density Memory Arrays** complying with Applied Materials Hackathon specifications. Compares **Classical CV** against **PyTorch ML (SEMRestorationUNet)**.

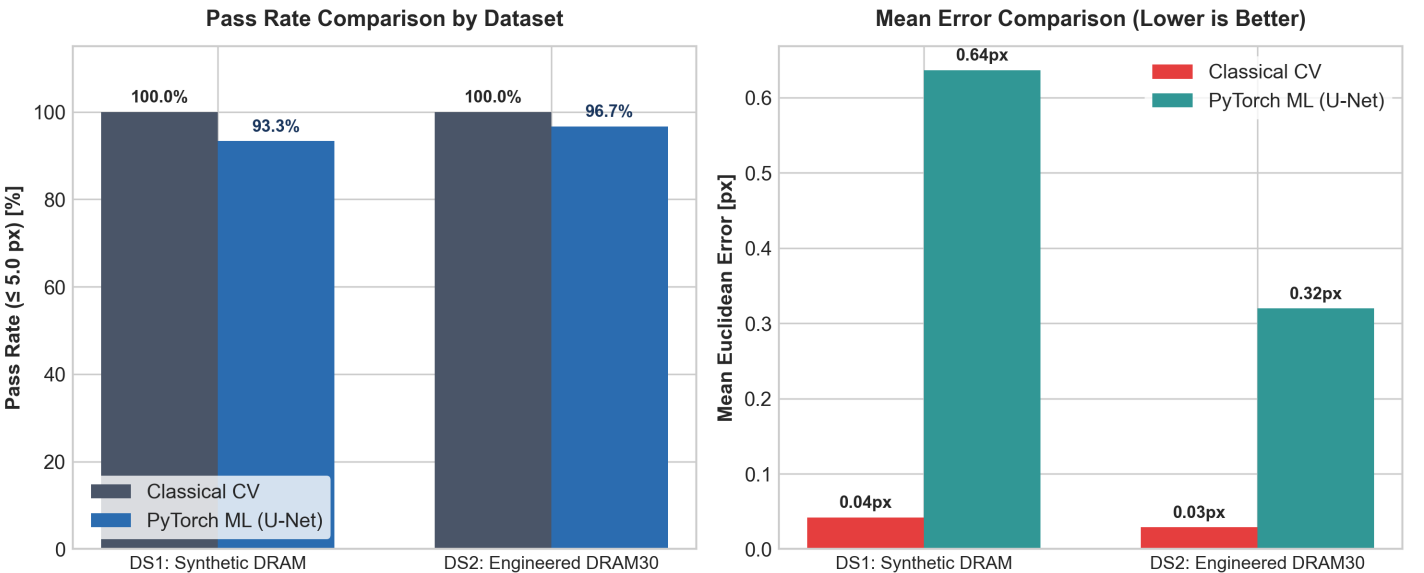
Key Performance Highlights (live-computed from benchmark run):

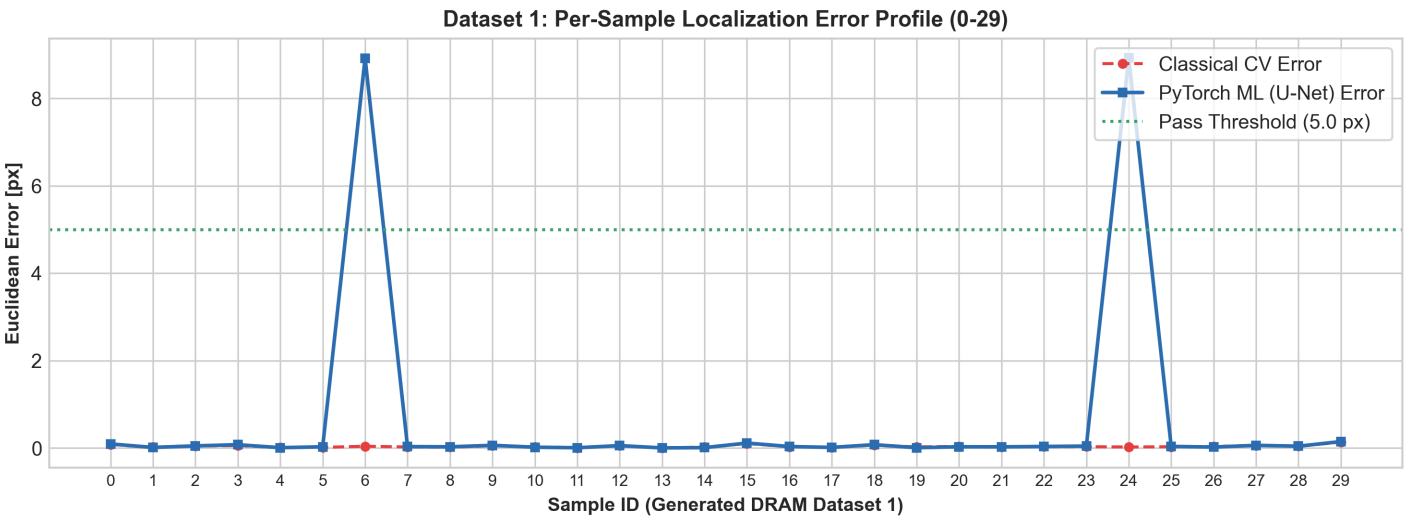
- Dataset 2 (Engineered DRAM30):** Classical CV achieves **100.0% Pass Rate @ ≤5px** with a median error of **0.03 px** and worst-case error of **0.08 px**.
- Dataset 1 (Synthetic DRAM):** Classical CV: 100.0% → PyTorch ML: 93.3% (vs by 6.7pp); mean error delta = -0.59 px.

1. Multi-Threshold Performance Breakdown (1px, 2px, 4px, 5px, Sub-Pixel)

Dataset	Pipeline Model	≤5px	≤4px	≤2px	≤1px	Sub-px (≤0.5)	Mean Err	Median	Max (Worst)	Runtime
DS1: Synthetic	Classical CV	100.0%	100.0%	100.0%	100.0%	100.0%	0.0px	0.0px	0.1px	2361ms
DS1: Synthetic	PyTorch ML	93.3%	93.3%	93.3%	93.3%	93.3%	0.6px	0.0px	8.9px	4665ms
DS2: DRAM30	Classical CV	100.0%	100.0%	100.0%	100.0%	100.0%	0.03px	0.03px	0.08px	5500ms
DS2: DRAM30	PyTorch ML	96.7%	96.7%	96.7%	96.7%	96.7%	0.32px	0.03px	8.68px	7883ms

2. Visual Analytics





3. Dataset 1: Synthetic DRAM SEM Pairs (Per-Sample Data)

30 Synthetic DRAM SEM Image Pairs evaluated with Scale Ratio and Rotation variations.

ID	Scale	Rot	True (X, Y)	CV Pred (X, Y)	CV Err	CV Stat	ML Pred (X, Y)	ML Err	ML Stat
00	9.0	-2.0°	(390.3, 567.9)	(390.4, 568.0)	0.08px	PASS	(390.4, 568.0)	0.10px	PASS
01	10.0	-1.0°	(349.9, 592.5)	(349.9, 592.5)	0.02px	PASS	(349.9, 592.5)	0.02px	PASS
02	11.0	+0.0°	(583.1, 596.9)	(583.1, 596.9)	0.05px	PASS	(583.1, 596.9)	0.05px	PASS
03	9.0	+1.0°	(561.3, 584.3)	(561.2, 584.2)	0.06px	PASS	(561.2, 584.2)	0.08px	PASS
04	10.0	+2.0°	(568.5, 486.3)	(568.5, 486.3)	0.02px	PASS	(568.5, 486.3)	0.01px	PASS
05	11.0	-2.0°	(514.5, 397.1)	(514.5, 397.1)	0.02px	PASS	(514.5, 397.1)	0.03px	PASS
06	9.0	-1.0°	(571.2, 552.9)	(571.2, 553.0)	0.04px	PASS	(580.1, 553.1)	8.92px	FAIL
07	10.0	+0.0°	(594.1, 512.7)	(594.1, 512.7)	0.03px	PASS	(594.1, 512.7)	0.03px	PASS
08	11.0	+1.0°	(553.4, 605.4)	(553.4, 605.4)	0.02px	PASS	(553.4, 605.4)	0.03px	PASS
09	9.0	+2.0°	(426.1, 364.9)	(426.1, 364.9)	0.05px	PASS	(426.1, 364.9)	0.06px	PASS
10	10.0	-2.0°	(651.8, 552.1)	(651.8, 552.1)	0.03px	PASS	(651.8, 552.1)	0.02px	PASS
11	11.0	-1.0°	(676.8, 441.0)	(676.8, 441.0)	0.01px	PASS	(676.8, 441.0)	0.01px	PASS
12	9.0	+0.0°	(495.1, 419.9)	(495.1, 419.9)	0.06px	PASS	(495.1, 419.9)	0.06px	PASS
13	10.0	+1.0°	(488.6, 566.7)	(488.6, 566.7)	0.01px	PASS	(488.6, 566.7)	0.01px	PASS
14	11.0	+2.0°	(527.6, 565.5)	(527.6, 565.5)	0.02px	PASS	(527.6, 565.5)	0.01px	PASS
15	9.0	-2.0°	(476.2, 535.6)	(476.1, 535.7)	0.10px	PASS	(476.1, 535.7)	0.12px	PASS
16	10.0	-1.0°	(427.8, 508.1)	(427.9, 508.1)	0.03px	PASS	(427.9, 508.1)	0.04px	PASS
17	11.0	+0.0°	(620.8, 276.7)	(620.8, 276.7)	0.02px	PASS	(620.8, 276.7)	0.02px	PASS
18	9.0	+1.0°	(556.0, 490.8)	(556.0, 490.9)	0.07px	PASS	(556.0, 490.9)	0.08px	PASS
19	10.0	+2.0°	(426.0, 443.7)	(426.0, 443.8)	0.03px	PASS	(426.0, 443.8)	0.01px	PASS
20	11.0	-2.0°	(492.6, 546.1)	(492.5, 546.2)	0.04px	PASS	(492.5, 546.2)	0.03px	PASS
21	9.0	-1.0°	(385.9, 508.7)	(386.0, 508.7)	0.03px	PASS	(386.0, 508.7)	0.03px	PASS
22	10.0	+0.0°	(354.3, 435.5)	(354.3, 435.5)	0.04px	PASS	(354.3, 435.5)	0.04px	PASS
23	11.0	+1.0°	(442.6, 482.7)	(442.6, 482.7)	0.03px	PASS	(442.6, 482.7)	0.05px	PASS
24	9.0	+2.0°	(664.0, 563.2)	(664.0, 563.2)	0.03px	PASS	(672.9, 562.9)	8.93px	FAIL
25	10.0	-2.0°	(614.4, 472.8)	(614.4, 472.8)	0.03px	PASS	(614.4, 472.8)	0.04px	PASS
26	11.0	-1.0°	(341.0, 530.0)	(341.0, 530.1)	0.04px	PASS	(341.0, 530.1)	0.02px	PASS
27	9.0	+0.0°	(562.8, 433.4)	(562.8, 433.4)	0.06px	PASS	(562.8, 433.4)	0.06px	PASS
28	10.0	+1.0°	(414.8, 315.1)	(414.8, 315.0)	0.04px	PASS	(414.8, 315.0)	0.04px	PASS
29	11.0	+2.0°	(561.8, 498.0)	(561.8, 498.2)	0.14px	PASS	(561.8, 498.2)	0.15px	PASS

4. Dataset 2: Engineered DRAM_30 Benchmark (Per-Sample Data)

30 Engineered DRAM Wafer Test Pairs evaluating baseline calibration and sub-pixel accuracy.

Pair	Tier	Scale	Rot	True (X, Y)	CV Pred (X, Y)	CV Err	CV Stat	ML Pred (X, Y)	ML Err	ML Stat
01	DRAM	10.00	+0.0°	(500.0, 500.0)	(500.0, 500.0)	0.00px	PASS	(500.0, 500.0)	0.00px	PASS
02	DRAM	10.00	+0.0°	(480.0, 520.0)	(480.0, 520.0)	0.00px	PASS	(480.0, 520.0)	0.00px	PASS
03	DRAM	9.95	+0.1°	(529.9, 469.9)	(530.0, 469.9)	0.06px	PASS	(530.0, 469.9)	0.07px	PASS
04	DRAM	10.05	-0.1°	(459.9, 539.9)	(460.0, 539.9)	0.03px	PASS	(460.0, 539.9)	0.03px	PASS
05	DRAM	10.00	+0.0°	(510.0, 490.0)	(510.0, 490.0)	0.00px	PASS	(510.0, 490.0)	0.00px	PASS
06	DRAM	9.10	+0.0°	(500.0, 500.0)	(500.0, 500.0)	0.02px	PASS	(500.0, 500.0)	0.01px	PASS
07	DRAM	9.50	+0.0°	(470.0, 530.0)	(470.0, 530.0)	0.02px	PASS	(470.0, 530.0)	0.02px	PASS
08	DRAM	10.00	+0.0°	(520.0, 480.0)	(520.0, 480.0)	0.00px	PASS	(520.0, 480.0)	0.00px	PASS
09	DRAM	10.50	+0.0°	(450.0, 550.0)	(450.0, 550.0)	0.01px	PASS	(450.0, 550.0)	0.02px	PASS
10	DRAM	10.90	+0.0°	(530.0, 460.0)	(530.0, 460.0)	0.04px	PASS	(530.0, 460.0)	0.05px	PASS
11	DRAM	10.00	-1.8°	(500.0, 500.0)	(500.0, 500.0)	0.02px	PASS	(500.0, 500.0)	0.02px	PASS
12	DRAM	10.00	-1.0°	(479.6, 519.7)	(479.7, 519.7)	0.03px	PASS	(479.7, 519.7)	0.03px	PASS
13	DRAM	10.00	+0.8°	(519.7, 479.7)	(519.7, 479.8)	0.04px	PASS	(519.7, 479.8)	0.04px	PASS
14	DRAM	10.00	+1.5°	(461.1, 541.0)	(461.0, 541.0)	0.03px	PASS	(461.0, 541.0)	0.03px	PASS
15	DRAM	10.00	+2.0°	(538.6, 458.6)	(538.6, 458.7)	0.05px	PASS	(538.6, 458.7)	0.05px	PASS
16	DRAM	10.00	+0.0°	(500.0, 500.0)	(500.0, 500.0)	0.01px	PASS	(500.0, 500.0)	0.01px	PASS
17	DRAM	10.00	+0.0°	(470.0, 530.0)	(470.0, 530.0)	0.02px	PASS	(470.0, 530.0)	0.02px	PASS
18	DRAM	10.00	+0.0°	(530.0, 470.0)	(530.0, 470.0)	0.04px	PASS	(530.0, 470.0)	0.04px	PASS
19	DRAM	10.00	+0.0°	(450.0, 550.0)	(450.0, 550.0)	0.01px	PASS	(450.0, 550.0)	0.01px	PASS
20	DRAM	10.00	+0.0°	(520.0, 480.0)	(520.0, 480.0)	0.02px	PASS	(520.0, 480.0)	0.02px	PASS
21	DRAM	9.20	+1.6°	(354.3, 654.1)	(354.3, 654.2)	0.04px	PASS	(345.6, 654.4)	8.68px	FAIL
22	DRAM	9.60	-1.4°	(653.6, 353.7)	(653.6, 353.7)	0.03px	PASS	(653.6, 353.7)	0.03px	PASS
23	DRAM	10.40	+1.8°	(306.4, 706.2)	(306.4, 706.2)	0.04px	PASS	(306.4, 706.2)	0.04px	PASS
24	DRAM	10.80	-1.7°	(705.8, 306.0)	(705.8, 306.1)	0.03px	PASS	(705.8, 306.1)	0.03px	PASS
25	DRAM	9.40	+1.2°	(402.1, 602.1)	(402.1, 602.1)	0.02px	PASS	(402.1, 602.1)	0.02px	PASS
26	DRAM	9.10	+1.9°	(210.9, 229.4)	(210.9, 229.4)	0.03px	PASS	(210.9, 229.4)	0.04px	PASS
27	DRAM	10.90	-1.9°	(770.5, 789.1)	(770.6, 789.1)	0.08px	PASS	(770.6, 789.1)	0.09px	PASS
28	DRAM	9.20	-2.0°	(241.4, 741.1)	(241.4, 741.1)	0.05px	PASS	(241.5, 741.1)	0.09px	PASS
29	DRAM	10.80	+2.0°	(741.1, 241.4)	(741.1, 241.4)	0.03px	PASS	(741.1, 241.4)	0.04px	PASS
30	DRAM	9.10	+1.9°	(210.1, 809.8)	(210.1, 809.8)	0.04px	PASS	(210.1, 809.8)	0.04px	PASS